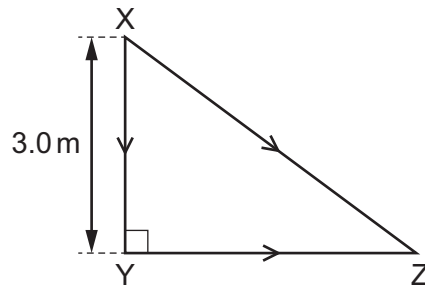


- 17 An object is falling in a uniform gravitational field.

Which two quantities are sufficient to calculate the change in gravitational potential energy?

- A mass and acceleration of free fall
- B mass and change in vertical displacement
- C weight and acceleration of free fall
- D weight and change in vertical displacement

- 18 The points X, Y and Z are on a rough, horizontal surface.



A box P is pushed across the surface from X to Y and then from Y to Z.

The distance from X to Y is 3.0 m. The work done against the frictional force in moving the box from X to Y is 150 J.

The work done against the frictional force in moving the box from Y to Z is 200 J.

An identical box Q is pushed in a straight line from X to Z.

The magnitude of the frictional force between the boxes and the surface is constant.

How much extra work is done against the frictional force in moving P than Q?

- A 100 J
- B 250 J
- C 350 J
- D 600 J

- 19 The momentum of a car of mass m increases from p_1 to p_2 .

What is the increase in the kinetic energy of the car?

- A $\frac{(p_2^2 - p_1^2)}{2m}$
- B $\frac{(p_2 - p_1)^2}{2m}$
- C $\frac{p_2 - p_1}{2m}$
- D $\frac{p_1 - p_2}{2m}$